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Nonavalent HPV vaccine to prevent recurrent anal or vulvar high-grade squamous intraepithelial lesions (VIVA trial): A randomized, double-blind, placebo-controlled trial

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Abstract

The nonavalent human papillomavirus (9vHPV) vaccine protects against HPV infection and high-grade squamous intraepithelial lesions (HSIL) when administered prior to exposure, but evidence supporting its potential therapeutic benefit has been inconsistent. This randomized, double-blind, placebo-controlled trial evaluated whether the 9vHPV vaccine reduces recurrence of HSIL or HPV persistence in 27–69-year-old persons previously treated for anal or vulvar HSIL. Participants were HSIL-free at enrollment and received 9vHPV or placebo at months 0, 2, and 6. High-resolution anoscopy or vulvoscopy was performed at months 18 and 36, and anal or vulvar swabs were collected at months 0, 18, 24, and 36 for HPV DNA detection. The primary endpoints were HSIL recurrence and HPV persistence (≥2 consecutive positive swabs for the same 9vHPV-type). Of 185 participants included in the intent-to-treat analysis, 91 received vaccine and

Abbreviations: ANCHOR, Anal Cancer-HSIL Outcomes Research trial; COVID-19, Coronavirus Disease 2019; CSS, Cancer Surveillance System; DNA, deoxyribonucleic acid; DSMB, Data and Safety Monitoring Board; HIV, human immunodeficiency virus; HPV, Human Papillomavirus; HRA, high-resolution anoscopy; hrHPV, high-risk human papillomavirus; HSIL, high-grade squamous intraepithelial lesion; IRB, Institutional Review Board; ITT, intent-to-treat; LSIL, low-grade squamous intraepithelial lesion; NCT, National Clinical Trial; SEER, Surveillance, Epidemiology, and End Results Program; UAB, University of Alabama Birmingham; UW, University of Washington; VRC, Virology Research Clinic; 9vHPV, nonavalent HPV vaccine.

Drs. Madeleine and Wald contributed equally to this manuscript.

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94 received placebo. The DSMB recommended early termination for futility. The 9vHPV vaccine was not more effective than placebo in preventing recurrent HSIL, with 16 HSIL among 9vHPV recipients versus 21 HSIL in placebo recipients (incidence 8.1 vs. 10.1/100 person-years; $p = .54$). HPV persistence was 21% in vaccine versus 31% in placebo recipients ($p = .20$). The 9vHPV vaccine delivered after treatment of anal or vulvar HSIL did not reduce HSIL recurrence or HPV detection. Our study underlines the importance of HPV vaccine administration prior to HPV exposure and the need for novel treatments for HSIL with high recurrence potential.

KEYWORDS
anal HSIL, HPV vaccination, human papillomavirus, vulvar HSIL

What's New?

Anal and vulvar high-grade squamous intraepithelial lesions (HSILs) frequently recur following treatment. Additional genital and anal procedures, however, can be distressing for patients and are potentially disfiguring. This trial assessed whether the 9-valent human papillomavirus (9vHPV) vaccine reduces HSIL recurrence risk or HPV persistence in persons treated for anal or vulvar HSIL. Results show that the 9vHPV vaccine did not reduce HSIL recurrence. In addition, the persistence of HPV genotypes was similar between vaccine versus placebo recipients. The findings highlight the importance of HPV vaccination in HPV-naïve persons and the need for novel HSIL treatment strategies.

1 | INTRODUCTION

The incidence of anal and vulvar cancer has risen significantly in recent decades, with more than 17,000 cases diagnosed annually in the United States.[1] Persistent infection with high-risk human papillomavirus (hrHPV) is the most critical risk factor identified in the development of anal and vulvar high-grade squamous intraepithelial lesions (HSIL), an epithelial abnormality that precedes most invasive anal and vulvar cancer.[2–4] The ANCHOR study demonstrated that treatment of anal HSIL decreases the risk of anal cancer leading to clinical guidelines for screening and treatment.[5–7] Treatment is also recommended for women with vulvar HSIL to interrupt progression to invasive cancer.[8] However, up to 50% of patients with HSIL develop recurrence of anal and vulvar precancers despite treatment, necessitating additional therapies that can lead to substantial physical and psychosocial morbidity.[9–12]

Licensed multivalent HPV vaccines provide nearly universal protection against HPV types included in the vaccine when administered to HPV-uninfected persons.[13] By preventing infection, the vaccine prevents HPV-related cervical disease, including cancer; prevention of anal and vulvar intraepithelial lesions has also been demonstrated.[14–16] Findings from studies evaluating a benefit of the vaccine in persons with prior HPV infection and pre-existing HPV-associated disease have varied. Multiple studies found that adjuvant HPV vaccination reduces lesion recurrence in immunocompetent patients surgically treated for cervical HSIL.[17,18] Observational studies also suggested that the risk of subsequent HSIL was significantly lower in persons vaccinated following treatment of anal and vulvar HSIL.[15,19,20] The putative mechanism for preventing recurrent disease is thought to be the overwhelming antibody response after vaccination that reduces the risk of reinfection or new infections at the basal cell layer after the prevalent disease is treated. However, recent randomized clinical trials among persons with HIV (PWH) did not support HPV vaccination to prevent recurrent anal HSIL.[21,22] To our knowledge, no prior randomized clinical trial has assessed the use of HPV vaccination to reduce recurrence in patients with a history of vulvar HSIL.

To expand the clinical evidence for use of the prophylactic HPV vaccine in patients with a history of HSIL, we designed the HPV Vaccine to Interrupt Progression of Vulvar and Anal Intraepithelial Lesions trial (VIVA trial) to test whether the nonavalent HPV vaccine (Gardasil®9) reduces HSIL recurrence in previously unvaccinated persons treated for anal or vulvar HSIL. Furthermore, we assessed persistence of hrHPV among vaccine recipients compared with placebo recipients.

2 | MATERIALS AND METHODS

2.1 | Trial design, setting, and oversight

The VIVA Trial was a randomized, double-blind, placebo-controlled trial conducted from August 2017 to December 2022. The study was initiated at the University of Washington and Fred Hutchinson Cancer Center, Seattle, Washington, and subsequently expanded to the University of Alabama at Birmingham, Birmingham, Alabama in January 2021. Detailed methods of the trial protocol were published previously[23] and additional details are provided in the Supplementary Materials.

2.2 | Trial participants

Potential participants were identified through the Fred Hutchinson Cancer Surveillance System (CSS, a Surveillance, Epidemiology, and End Results [SEER] cancer registry in western Washington) and referral by local healthcare providers. The registry collects epidemiologic information on anal and vulvar HSIL and cancer as part of its mandate to identify new cancer cases within 45 days of diagnosis. Participants were eligible for randomization if they were 27–69 years of age, had prior histologically confirmed anal or vulvar HSIL, were free of HSIL at screening by high-resolution anoscopy (HRA) or vulvoscopy, were treated with antiretrovirals for at least 6 months if they had HIV infection, and were able to provide informed consent. Key exclusion criteria were prior HPV vaccination, pregnancy, and history of HPV-related cancer. Detailed inclusion and exclusion criteria are provided in the Supplementary Materials.

2.3 | Screening and enrollment

After written informed consent, potential participants provided demographic and medical history. Persons with unknown HIV status were screened for HIV infection, and persons of childbearing potential had a pregnancy test. A physical examination and HRA or vulvoscopy, depending on the anatomic site of the qualifying lesion, were performed. Directed biopsies were obtained of anal or vulvar lesions found on exam to rule out HSIL. Participants with no clinical evidence of HSIL or with a biopsy without HSIL on exam were eligible for enrollment. Anal or vulvar samples were collected for HPV DNA genotyping and placed into ThinPrep® vials.

2.4 | Randomization and vaccination

Participants were randomly assigned 1:1 to 9vHPV vaccine or placebo (normal saline), using dynamic randomization[24] designed to balance assignment of the participants by anatomic site of HSIL (anus or vulva), HIV status (positive or negative), and time since HSIL diagnosis (<12 months or ≥12 months). Trial personnel and participants were masked to study vaccine assignment; however, an unblinded provider administered the vaccine in a masked syringe at 0, 2, and 6 months. Participants were observed for 15 min following receipt of each vaccination and received a diary to record side effects for 5 days following each vaccination.

2.5 | Study follow-up and procedures

Participants returned for follow-up visits at 7, 18, 24, and 36 months for an interim medical history and physical exam, and blood draw. At study months 18, 24, and 36, vulvar or anal swabs were obtained. Participants underwent HRA or vulvoscopy at months 18 and 36; lesions suspicious for HSIL were biopsied.

Follow-up telephone visits were conducted at 12 and 42 months to collect updated health information and any anogenital concerns. Participants were permitted to continue routine follow-up with their primary care providers, including undergoing HRA or vulvoscopy if clinically indicated. Pathology results of anogenital biopsies and medical records of HSIL treatment conducted outside the study were retrieved and abstracted for the study during the course of the trial.

2.6 | Laboratory methods

HPV genotyping from anal and vulvar samples was performed using a Luminex assay, a liquid bead microarray real-time polymerase chain reaction (PCR)-based assay, at the University of Washington, Department of Laboratory Medicine and Pathology.[25] This assay detects 37 HPV genotypes, including those covered by the 9vHPV vaccine (HPV 6, 11, 16, 18, 31, 33, 45, 52, 58). The Luminex assay result is reported as positive or negative for the presence of each of the 37 HPV types if detected in the sample. Biopsy specimens obtained throughout the study at both sites were reviewed by a single pathologist at the University of Washington (VG).

2.7 | Outcomes and assessment

The initial primary endpoint was time to recurrence of anal or vulvar histology-confirmed HSIL ("histological endpoint") among participants who received the 9vHPV vaccine compared to placebo recipients, which was assessed in the intent-to-treat population (ITT). The ITT cohort included all randomized participants who received at least one dose of vaccine or placebo and who remained HSIL-free at least 30 days. After a protocol amendment due to lower-than-expected enrollment rate and study pause during the early COVID-19 pandemic (Table S1), the trial primary endpoint was changed to HPV persistence defined as detection of the same high-risk 9vHPV vaccine type (HPV 16, 18, 31, 33, 45, 52, 58) in ≥2 consecutive anal or vulvar swab samples ("virologic endpoint") detected during the study follow-up. Persistent HPV infection was chosen as the virologic endpoint because it is a validated biomarker of histologic progression, allowing for a meaningful assessment even with a smaller sample size. The rationale for defining persistence as having two consecutive positive swabs for the same vaccine HPV genotype was to minimize the inclusion of transient HPV detection. The decision to begin event accrual at day 30 was made to ensure that the humoral response to the vaccine had sufficient time to develop and impact HSIL recurrence, as antibodies peak by week 4. Also, participants could follow up with their providers and potentially have HSIL detected outside of the study earlier than the scheduled visits at months 18 and 24, during which HRA and vulvoscopy exams were conducted within the study. Similarly, positive HPV PCR results at baseline were excluded to ensure that the analysis reflected the vaccine's impact on recurrent, rather than prevalent infections.

Safety was assessed in all participants who had received at least one dose of the 9vHPV vaccine or placebo. Local and systemic adverse events were collected at the time of each injection and for five subsequent days. Serious adverse events and severe vaccine-related adverse events were recorded through the entire study period.

2.8 | Statistical analyses

The sample size for the initial histological endpoint assumed a 30% incidence of HSIL recurrence over 3 years among those not vaccinated, and a reduction of the hazard of recurrence by 50%, to a 16% risk of HSIL recurrence over 3 years.[20,26] Enrollment of 345 participants was estimated to achieve 80% power with a two-sided 0.05 level of significance, allowing for 10% dropout. For the revised virologic endpoint, in the context of under-recruitment, a sample size of 230 participants was predicted to have 90% power to detect a 50% decrease in persistent vaccine-type HPV infection, from approximately 40% in the control group to 20% in the experimental group and a two-sided 0.05 level of significance. This higher power estimate was based on the higher prevalence of the virologic endpoint and the sample size already accrued at the time of the analysis plan modification. This higher power estimate was based on the higher prevalence of the virologic endpoint and the sample size already accrued at the time of the analysis plan modification.

To evaluate safety, two planned interim analyses were conducted by a Data Safety Monitoring Board (DSMB) after the initial 50 participants completed two vaccinations and again after 150 participants completed their Month 6 visit. Prompted by slow accrual, a third unplanned interim analysis of the primary outcome was performed to assess the futility of continuing the study. Using the proportion of persons accrued to date, we applied sequential stopping rules, using O'Brien-Fleming boundaries, to determine the futility threshold of detecting a significant difference between groups with the data accrued before full enrollment (See Supplementary Materials and Figure S1 for additional details).[27]

Kaplan–Meier methods and log-rank tests were used to evaluate if there was an important difference in the recurrence of HSIL among 9vHPV vaccine versus placebo recipients. The chi-square test was used to compare HPV persistence in the vaccine compared to the placebo group. For secondary analyses, we assessed vaccine efficacy in subsets of participants including the per-protocol population. It included participants who completed all three vaccine doses and did not have HSIL within 30 days of the last dose. Other subset analyses explored differences by randomization stratification factors including anatomic site (anal versus vulvar), HIV status (positive versus negative), and time since HSIL diagnosis (<12 months versus ≥12 months).

Safety was assessed by comparing adverse events between study groups. All analyses were performed using Stata/SE 18.0.

3 | RESULTS

3.1 | Participants

Between August 1, 2017, and December 31, 2022, 1855 persons were identified as potentially eligible for the study and 1102 completed the pre-screening call; 256 attended a screening visit in the clinic (Figure 1 and Table S2), with 59 (23%) failing screening. Among 149 potential participants with a history of anal HSIL who were screened, 33 (22.1%) had active anal HSIL, 9 (6.0%) failed for other reasons, and 3 (2.0%) refused participation. Among 104 potential participants with a history of vulvar HSIL who attended a screening visit, 13 (12.5%) were found to have active vulvar HSIL, 4 (3.8%) failed screening for other reasons, and 3 (2.8%) refused participation. A total of 188 participants met the eligibility criteria and underwent randomization, 93 to the 9vHPV vaccine group and 95 to the placebo group. The median length of follow-up was 25 months, and 185 participants had at least 30 days follow-up after the initial dose of vaccine and were included in the intention-to-treat analysis. A total of 162 (86.2%) participants returned for their scheduled visits and 97 participants completed the last visit. Of the 1415 expected study visits among enrolled participants, 1314 were completed (92.9%).

The baseline demographics, clinical characteristics, and randomization strata were balanced between the 9vHPV vaccine and placebo groups (Table 1 and S3). At enrollment, the median age of study participants was 55 years; 99 (53.5%) were women and 145 (78.4%) were White. A total of 103 (55.7%) participants had a history of anal HSIL as their qualifying anatomical site and 82 (44.3%) had a history of vulvar HSIL. A third of the participants were living with HIV (38.4%) and most (77.3%) had their initial HSIL lesion diagnosed ≥12 months before study start. Among male participants, 68 (76.4%) were living with HIV. In contrast, only 2 of 99 women (2%) were living with HIV.

Of 185 anal ($n = 103$) and vulvar ($n = 82$) swab samples obtained at baseline, one anal swab was excluded from the analyses. Twenty-five (13.5%) participants had one additional swab, and 117 (63.2%) had two or more available swabs after the baseline swab. The baseline distribution of HPV detection in anal and vulvar swabs is shown in Table 2. At least one high-risk 9vHPV genotype was detected in most participants, with a higher prevalence of HPV detected in anal (80.4%) swabs compared to vulvar (46.3%) swabs. Non-vaccine high-risk HPV genotypes were detected in 22% of anal swabs and 17% of vulvar swabs.

3.2 | Study outcomes

Thirty-seven participants with recurrent HSIL were identified during 406 person-years of follow-up; 16 participants in the vaccine group and 21 in the placebo group. The incidence of recurrent HSIL among participants in the vaccine group was 8.1 per 100 person-years (95% CI, 5.0–13.3) compared with 10.1 per 100 person-years (95% CI, 6.6–15.4) in the placebo group ($p = .54$, log-rank test) (Table 3A and Figure 2). Persistence of the virological endpoint occurred in 31 of

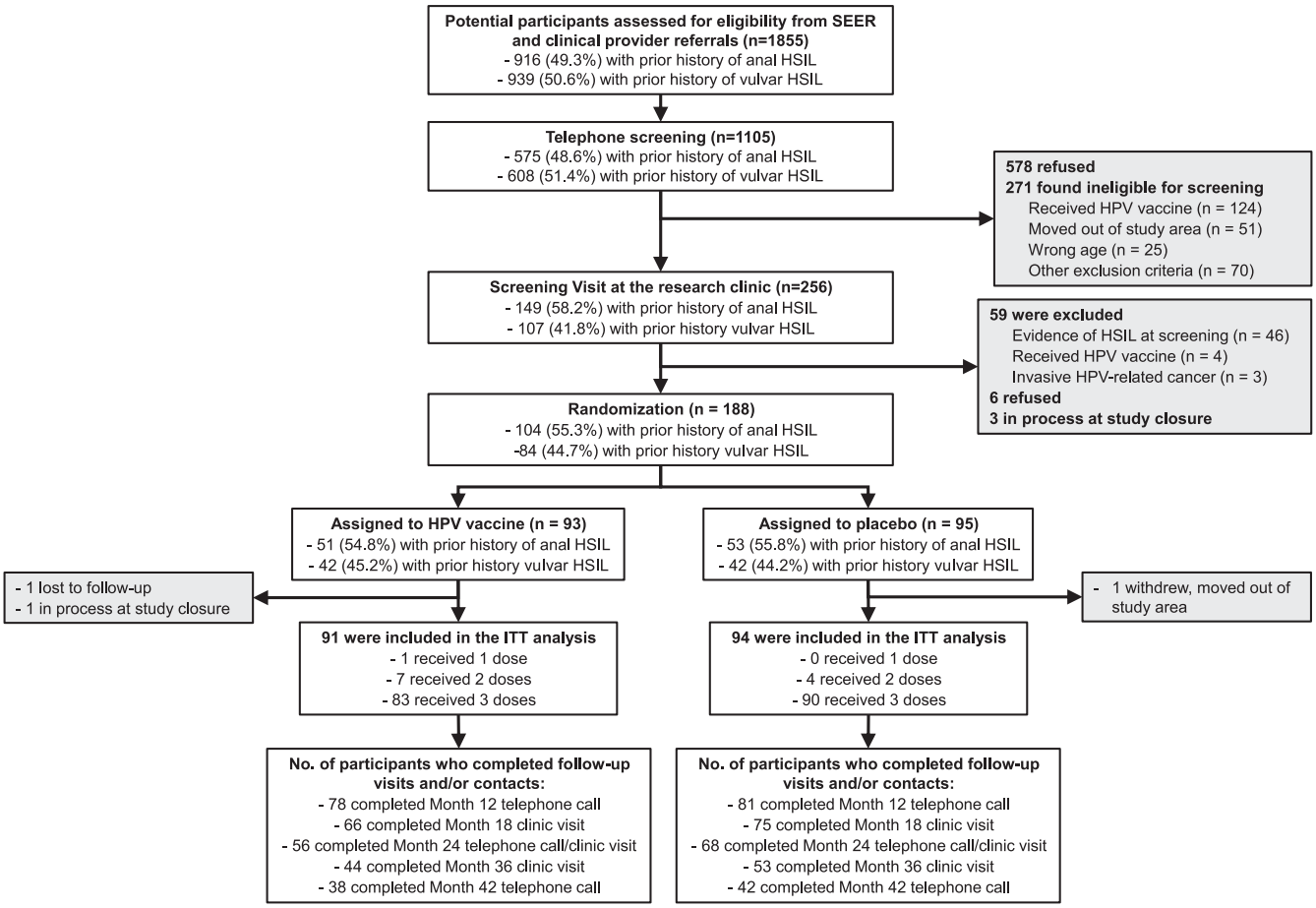


FIGURE 1 CONSORT diagram for VIVA trial.

117 (26%) participants included in the analysis. Persistent HPV was observed in 21% of those in the vaccine group compared to 31% in the placebo group ($p = .20$) (Table 3B).

In secondary analyses of the ITT population, we assessed incidence of recurrent HSIL and HPV persistence in subsets grouped by anatomic site, HIV status, and time since diagnosis of the qualifying lesion to demonstrate the variation in efficacy across baseline strata (Figures S2–S4; Tables S4 and S5). However, these analyses were underpowered due to the small sample sizes. The incidence of anal HSIL was 9.7 per 100 person-years in the vaccine group versus 13.5 per 100 person-years in the placebo group ($p = .32$). Among participants with a history of vulvar HSIL, the incidence of HSIL was 6.0 per 100 person-years in the vaccine group versus 6.1 per 100 person-years in the placebo arm ($p = .99$). In participants with HIV, the incidence of HSIL was 11.0 per 100 person-years among vaccine recipients versus 16.8 among placebo recipients ($p = 0.28$). In those without HIV infection, the incidence of HSIL was 6.4 per 100 person-years in the vaccine group versus 6.6 per 100 person-years in the placebo group ($p = 0.99$). In the subset of participants with an initial diagnosis of HSIL less than 12 months from the time of enrollment, the incidence of HSIL was 7.1 per 100 person-years in the vaccine group versus 11.2 among placebo recipients ($p = .56$). In those with an initial diagnosis of HSIL of 12 or more months, the incidence of HSIL was 8.4 per 100 person-years versus 9.7 per 100 person-years in the vaccine and placebo groups, respectively ($p = .71$). Overall, HPV persistence did not differ between vaccine and placebo recipients when stratified by anatomical site, HIV status, or time since HSIL diagnosis (Table S5). However, persistence of HPV types 31/33/52/28 was significantly lower among vaccine recipients with prior anal HSIL compared to placebo (6% vs. 27%, $p = .02$). Similarly, persistence of hrHPV vaccine types was lower in HIV-positive vaccine recipients than in HIV-positive placebo recipients (23% vs. 54%, $p = .03$). In the per-protocol population that included participants who received all 3 vaccine doses and did not have recurrent HSIL within 30 days of last dose, the incidence of recurrent HSIL was 6.2 per 100 person-years in the vaccine group compared to 7.9 per 100 person-years in the placebo group ($p = .54$) (Figure S5 and Table S4).

3.3 | Safety analyses

The most frequently reported vaccine-related reactogenicity were injection site tenderness and pain; these occurred more frequently among 9vHPV vaccine recipients (Table 4). Three severe adverse events (fatigue, collar bone throbbing, and toe throbbing), occurring in

TABLE 1 Baseline characteristics for the intent-to-treat (ITT) population[a].

	HPV vaccine		Placebo		Overall	
	(*n* = 91)		(*n* = 94)		(*n* = 185)	
	n	%	*n*	%	*n*	%
Median age at enrollment (years, IQR)	53	(47–61)	57	(51–63)	55	(48–62)
Gender						
Women	49	(53.8)	50	(53.2)	99	(53.5)
Men	41	(45.1)	44	(46.8)	85	(45.9)
Transgender women	1	(1.1)	0	(0.0)	1	(0.5)
Race						
American Indian or Alaska Native	0	(0.0)	1	(1.1)	1	(0.5)
Asian	2	(2.2)	2	(2.1)	4	(2.2)
Native Hawaiian or Pacific Islander	1	(1.1)	0	(0.0)	1	(0.5)
Black	9	(9.9)	6	(6.4)	15	(8.1)
White	70	(76.9)	75	(79.8)	145	(78.4)
Multiracial	5	(5.5)	7	(7.4)	12	(6.5)
Unknown	4	(4.4)	3	(3.2)	7	(3.8)
Ethnicity						
Not Hispanic	84	(92.3)	89	(94.7)	173	(93.5)
Hispanic	5	(5.5)	3	(3.2)	8	(4.3)
Unknown	2	(2.2)	2	(2.1)	4	(2.2)
Anatomic site of HSIL						
Anal	50	(54.9)	53	(56.4)	103	(55.7)
Vulvar	41	(45.1)	41	(43.6)	82	(44.3)
Time since qualifying HSIL diagnosis						
<12 months	19	(20.9)	23	(24.5)	42	(22.7)
12 or more months	72	(79.1)	71	(75.5)	143	(77.3)
HIV status						
Negative	59	(64.8)	55	(58.5)	114	(61.6)
Positive	32	(35.2)	39	(41.5)	71	(38.4)
Median CD4 count (IQR)[b]	583	(324–772)	675	(500–800)	670	(467–799)
Immunosuppressive condition[c]						
No	87	(95.6)	90	(95.7)	177	(95.7)
Yes	4	(4.4)	4	(4.3)	8	(4.3)
Smoking status						
Never	40	(44.0)	52	(55.3)	92	(49.7)
Former	31	(34.1)	24	(25.5)	55	(29.7)
Current	20	(22.0)	18	(19.1)	38	(20.5)
New sexual partner in the last year						
No	67	(73.6)	69	(73.4)	136	(73.5)
Yes	24	(26.4)	22	(23.4)	46	(24.9)
Missing	0	(0.0)	3	(3.2)	3	(1.6)

Abbreviations: HIV, human immunodeficiency virus; HPV, human papillomavirus; HSIL, high-grade squamous intraepithelial lesion; IQR, interquartile rage.
[a]ITT population: participants who received at least one vaccine and remained HSIL-free for at least 30 days after the first vaccine dose.
[b]Median CD4 count among participants with HIV. Unknown value for $n = 13$.
[c]Immunosuppressive conditions included: solid organ transplant, bone marrow transplant, systemic lupus erythematosus on immunosuppressive therapy, Sjögren syndrome on immunosuppressive therapy.

TABLE 2 Baseline human papillomavirus detection in anal and vulvar swabs.

	Anal						Vulvar						Overall					
	HPV vaccine		Placebo		Total		HPV vaccine		Placebo		Total		HPV vaccine		Placebo		Total	
	(*n* = 50)		(*n* = 53)		(*n* = 103)		(*n* = 41)		(*n* = 41)		(*n* = 82)		(*n* = 91)		(*n* = 94)		(*n* = 185)	
	n	%	*n*	%	*n*	%	*n*	%	*n*	%	*n*	%	*n*	%	*n*	%	*n*	%
Any hrHPV[a]																		
No	11	(22.4)	9	(17.0)	20	(19.6)	16	(39.0)	22	(53.7)	38	(46.3)	27	(30.0)	31	(33.0)	58	(31.5)
Yes	38	(77.6)	44	(83.0)	82	(80.4)	25	(61.0)	19	(46.3)	44	(53.7)	63	(70.0)	63	(67.0)	126	(68.5)
Missing	1		0		1								1		0		1	
HPV16																		
No	37	(75.5)	40	(75.5)	77	(75.5)	33	(80.5)	33	(80.5)	66	(80.5)	70	(77.8)	73	(77.7)	143	(77.7)
Yes	12	(24.5)	13	(24.5)	25	(24.5)	8	(19.5)	8	(19.5)	16	(19.5)	20	(22.2)	21	(22.3)	41	(22.3)
Missing	1		0		1								1		0		1	
HPV18/45																		
No	43	(87.8)	43	(81.1)	86	(84.3)	37	(90.2)	39	(95.1)	76	(92.7)	80	(88.9)	82	(87.2)	162	(88.0)
Yes	6	(12.2)	10	(18.9)	16	(15.7)	4	(9.8)	2	(4.9)	6	(7.3)	10	(11.1)	12	(12.8)	22	(12.0)
Missing	1		0		1								1		0		1	
HPV31/33/52/58																		
No	34	(69.4)	31	(58.5)	65	(63.7)	33	(80.5)	36	(87.8)	69	(84.1)	67	(74.4)	67	(71.3)	134	(72.8)
Yes	15	(30.6)	22	(41.5)	37	(36.3)	8	(19.5)	5	(12.2)	13	(15.9)	23	(25.6)	27	(28.7)	50	(27.2)
Missing	1		0		1								1		0		1	
Non-vaccine hrHPV[b]																		
No	38	(77.6)	41	(77.4)	79	(77.5)	34	(82.9)	34	(82.9)	68	(82.9)	72	(80.0)	75	(79.8)	147	(79.9)
Yes	11	(22.4)	12	(22.6)	23	(22.5)	7	(17.1)	7	(17.1)	14	(17.1)	18	(20.0)	19	(20.2)	37	(20.1)
Missing	1		0		1								1		0		1	
Other HPV[c]																		
No	25	(51.0)	19	(35.8)	44	(43.1)	27	(65.9)	31	(75.6)	58	(70.7)	52	(57.8)	50	(53.2)	102	(55.4)
Yes	24	(49.0)	34	(64.2)	58	(56.9)	14	(34.1)	10	(24.4)	24	(29.3)	38	(42.2)	44	(46.8)	82	(44.6)
Missing	1		0		1								1		0		1	
HPV6/11																		
No	39	(79.6)	45	(86.5)	84	(83.2)	41	(100.0)	38	(92.7)	79	(96.3)	80	(88.9)	83	(89.2)	163	(89.1)
Yes	10	(20.4)	7	(13.5)	17	(16.8)	0	(0.0)	3	(7.3)	3	(3.7)	10	(11.1)	10	(10.8)	20	(10.9)
Missing	1		1		2								1		1		2	

Abbreviations: HPV, human papillomavirus; hrHPV, high-risk human papillomavirus.

[a]HrHPV was defined as vaccine-type specific oncogenic HPV (HPV16, 18, 31, 33, 45, 52, 58).

[b]Additional hrHPV types identified at baseline but not present in the nonavalent vaccine included: 35, 39, 51, 56, 59.

[c]Additional low-risk HPV types identified at baseline included: 53, 55, 62, 64, 66, 67, 68, 69, 71, 73, 82, 83, 84, IS39.

a single placebo recipient, were considered related to the study vaccine (Table S6). There were 39 serious adverse events, 20 in the vaccine group and 19 in the placebo group (Table S6). Among 6 deaths during the trial period, 3 occurred in the vaccine group and 3 in the placebo group. None of the deaths was deemed related to the study vaccine or study trial procedures (Table S7).

3.4 | Futility analysis

At the time of the interim futility analysis, we had accrued 33 (46%) events of the total 72 events planned prior to changing the primary outcome. With 46% of HSIL events accrued, the O'Brien-Fleming threshold for concluding futility was $p > .60$. p-values for both the ITT and PP analyses were $p = .83$ and $p = .93$, respectively. We also computed a 6% conditional power to identify a significant difference in incidence of recurrent HSIL by study arm. Based on the accrued events, there was a 94% chance of *not* finding a significant difference after completing enrollment and following the original cohort size of the histology endpoint ($n = 345$). This conditional power was confirmed by simulation. Furthermore, the computation was unrelated to the secondary virological outcome, and unrelated to the reduced sample size of the study (see Supplementary Materials and Figure S1 for additional details).

TABLE 3A The incidence of recurrent anal or vulvar HSIL (histologic endpoint) by study arm.

	Events	Person-years	Incidence per 100 person-years	p-value[a]
Intent-to-treat				
Total	37	406	9.1	
HPV vaccine	16	197	8.1	
Placebo	21	209	10.1	.54
Per-protocol				
Total	28	395	7.1	
HPV vaccine	12	193	6.2	
Placebo	16	203	7.9	.54

Abbreviations: HPV, human papillomavirus; HSIL, high-grade squamous intraepithelial lesion.
[a]Log-rank test vaccine versus placebo.

4 | DISCUSSION

In persons with a prior history of anal or vulvar HSIL, the 9vHPV vaccine administered after treatment of HSIL did not reduce the risk of HSIL recurrence. Furthermore, while the persistence of HPV genotypes did not significantly differ by receipt of the 9vHPV vaccine, this finding may be attributable to limited statistical power. The results were also consistent across subgroups defined by anatomic site, HIV status, and time since HSIL diagnosis, although the subgroup analyses were underpowered. Consistent with prior trials in younger cohorts, the 9vHPV vaccine was safe and well-tolerated in our study, which included an older population. Although very safe, evidence from the VIVA trial does not support the use of the 9vHPV vaccine as an adjunct to treatment to reduce anal and vulvar HSIL recurrence.

Observational studies previously suggested a potential therapeutic benefit against recurrence in previously unvaccinated persons treated for anal HSIL.[20,28] In a retrospective cohort study of HIV seronegative men treated for HSIL, the risk of HSIL recurrence was significantly lower in the HPV vaccinated group versus unvaccinated group (13.6% vs. 30.7%, HR 0.5).[20] However, consistent with our findings, more recent investigations have not supported HPV vaccination as adjuvant treatment to prevent anal HSIL or persistent HPV DNA

TABLE 3B HPV persistence (virological endpoint) by study arm.

HPV persistence identified[a]	HPV vaccine	Placebo	Total	p-value
Intent-to-treat	($n = 53$)	($n = 64$)	($n = 117$)	
Overall (hrHPV)[b]	11 (21%)	20 (31%)	31 (26%)	.20
HPV16	7 (13%)	9 (14%)	16 (14%)	.89
HPV18/45	3 (6%)	5 (8%)	8 (7%)	.65
/33/52/58	3 (6%)	11 (17%)	14 (12%)	.056
HPV35/39/51/56/59	4 (8%)	8 (13%)	12 (10%)	.38
Per-protocol	($n = 52$)	($n = 60$)	($n = 112$)	
Overall (hrHPV)[b]	11 (21%)	18 (30%)	29 (26%)	.29
HPV16	7 (13%)	8 (13%)	15 (13%)	.98
HPV18/45	3 (6%)	5 (8%)	8 (7%)	.60
HPV31/33/52/58	3 (6%)	10 (17%)	13 (12%)	.07
HPV35/39/51/56/59	4 (8%)	6 (10%)	10 (9%)	.67

Abbreviations: HPV, human papillomavirus, hrHPV, high-risk human papillomavirus.
[a]HPV persistence was defined as detection of the same hrHPV vaccine type in ≥2 consecutive anal or vulvar swab samples.
[b]HrHPV was defined as vaccine-type specific oncogenic HPV (HPV16, 18, 31, 33, 45, 52, 58).

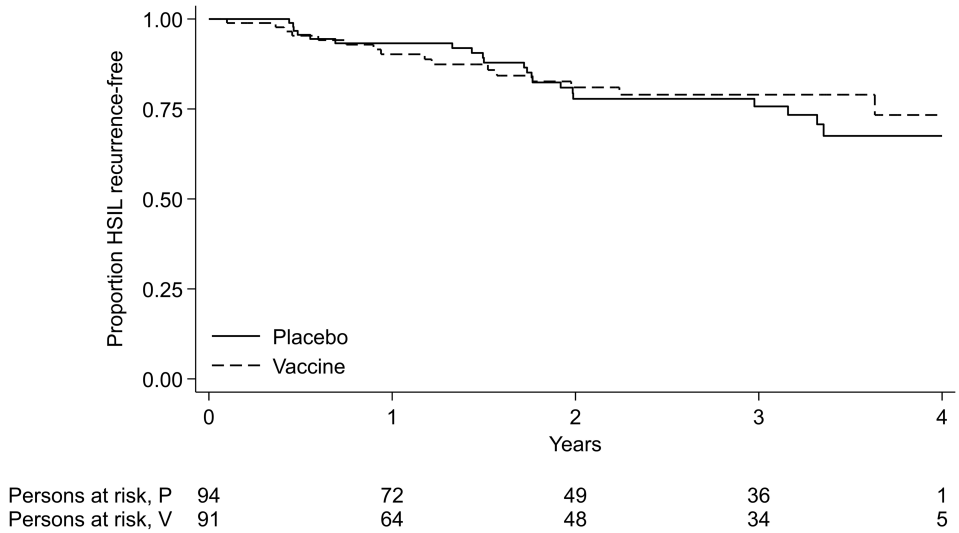


FIGURE 2 Kaplan–Meier curve for the histological primary endpoint among participants who received at least one vaccine and remained HSIL-free for at least 30 days after the first vaccine dose (intent-to-treat population).

TABLE 4 Solicited vaccine reactogenicity by study arm[a].

	HPV Vaccine (*n* = 91)		Placebo (*n* = 94)		Total (*n* = 185)	
	n	%	*n*	%	*n*	%
Symptom and signs						
Fever/Chills	6	(6.6)	2	(2.1)	8	(4.3)
Headache	24	(26.4)	22	(23.4)	46	(24.9)
Fatigue	23	(25.3)	25	(26.6)	48	(25.9)
Muscle aches	17	(18.7)	17	(18.1)	34	(18.4)
Injection site pain	33	(36.3)	17	(18.1)	50	(27.0)
Injection site tenderness	53	(58.2)	31	(33.0)	84	(45.4)
Injection site swelling	12	(13.2)	2	(2.1)	14	(7.6)
Medical attention/medication	5	(5.5)	11	(11.7)	16	(8.6)

[a]Symptoms reported at least once after any study vaccine dose.

detection in patients with HIV.[21,22] In a randomized trial conducted in the Netherlands among 126 men who have sex with men living with HIV with prior treatment of anal HSIL but free of disease at enrollment, the quadrivalent HPV (4vHPV) vaccine failed to prevent recurrent anal HSIL 18 months after vaccination.[21] Similarly, a randomized trial conducted in the US among 575 men and women with HIV showed no difference between the 4vHPV vaccine and control groups in persistent anal HPV and HSIL detection.[22] Although the mechanisms are not fully understood and prospective studies assessing clinical outcomes in PWH are limited, historical data suggest that a less robust immune response to HPV vaccines may explain the reduced clinical protection in this population, especially in those with low CD4 counts and high HIV viral loads.[29–31] Additionally, PWH have a higher prevalence and more extensive disease burden of anal squamous intraepithelial lesions, and recurrent lesions after treatment may result from the outgrowth of residual transformed tissue, where the prophylactic vaccine may not be effective.[32] Our clinical trial is the first to assess clinical outcomes with the 9vHPV vaccine, contributing new evidence to the existing literature that HPV vaccination with the 4- or 9-valent vaccine does not reduce anal HSIL recurrence in persons with and without HIV infection and those up to 69 years of age.

To our knowledge, this is the first randomized clinical trial assessing the use of HPV vaccination to reduce recurrence that included patients with a history of vulvar HSIL. Ghelardi et al. showed in a case–control study a 78.5% reduction in incident vulvar HSIL in persons vaccinated with the 4vHPV vaccine after surgical treatment.[33] However, all randomized trials and post-hoc analyses of randomized controlled trials investigating the use of adjuvant HPV vaccination to reduce HSIL recurrence in cis-women have been focused on cervical disease. In contrast to our findings, several studies support that the role of HPV vaccination is to reduce the risk of recurrent cervical HSIL in women treated with local excision.[34–37] One proposed mechanism is that surgical treatment of cervical dysplasia restores the cervical mucosa and the immunological milieu to a state similar to that of HPV naïve microenvironment, allowing the HPV vaccine to provide a prophylactic type of protection against new infections.[18] The more complete treatment of cervical HSIL may therefore account for the lower recurrence rates of cervical HSIL compared to vulvar or anal HSIL in women.[38–40]

Our trial is the first study evaluating the safety of the 9vHPV vaccine in those 27–69 years old with anal or vulvar HSIL. The 9vHPV vaccine was safe during the study follow-up of 42 months. Reactogenicity was less frequent than previously reported in younger cohorts from early licensure 9vHPV vaccine studies (e.g., injection-site pain 36.3% in VIVA trial versus 89.9% in younger persons).[41] We did not observe any safety concerns related to the 9vHPV vaccination, as indicated by a comparable proportion of participants reporting adverse events in both arms, confirming safety in this older age group up to 3.5 years of follow up. Long-term safety data for the 9vHPV vaccine in this group are currently lacking, primarily due to the vaccine not being licensed for persons older than 45 years of age.

We acknowledge several limitations of our trial. First, slower-than-anticipated accrual was an important limitation of the study. Despite having a large pool of potential participants, protocol adjustments, and the addition of a second study site, the target enrollment was not achieved. Moreover, participant recruitment was halted due to the prolonged suspension of our trial during the COVID-19 pandemic. Such challenges prompted the unplanned interim analysis to assess futility, resulting in the DSMB recommendation to stop the trial. Although stopping the trial early due to futility can lead to misestimation of the vaccine effect, the estimated predictive power with the accrued participants and histological endpoint suggested that a benefit from the vaccine was highly unlikely to be found even if the study were to continue to its planned completion. Second, the change in the primary endpoint from time to recurrence of histology-confirmed HSIL to HPV persistence could impact the evaluation of the 9vHPV vaccine's efficacy and the clinical relevance of the findings, as HPV persistence does not always correlate with lesion recurrence. Although this change redirected focus from direct clinical outcomes to biological markers and increased statistical power due to more frequent detection of viral events, we were mindful of this trade-off and ultimately assessed both outcomes. Furthermore, when we compared the predictive HSIL incidence of 11% over 1 year (30% over the 3 years), the observed incidence closely aligned with 8.1% and 10.1% in the two arms, respectively. Therefore, our expectations regarding the event rate were accurate. As the anticipated efficacy was determined by what we thought could drive interest in the practice of vaccination, the futility analysis was performed on the clinical endpoint, and it was conclusive. Since the futility analysis was conducted based on the histological outcome, it is possible that the limited statistical power of the study contributed to the non-significant results observed for the virologic endpoint. Third, although most anal and vulvar HSIL are associated with high-risk HPV types,[42] predominantly HPV16, the HPV status of the HSIL lesions in biopsy samples was not examined in our cohort and we were not able to ascertain if recurrent HSIL was caused by the same or a different HPV type. However, we used a central pathology review to limit observer variability for the histology endpoint. Fourth, we have not performed anal cytology in our participants with anal HSIL as qualifying lesions, which is a useful quality control measure for HRA as it can point to discrepancies when lesions are missed during the exam or when sampling error occurs. Nonetheless, HRA exam is the gold standard for the diagnosis of HSIL and it was performed by two experienced providers certified to perform HRA by the ANCHOR Quality Assessment Committee and the AIDS Malignancy Consortium. Fifth, regarding vulvar swab collections, obtaining adequate vulvar samples for HPV DNA genotyping presents unique challenges due to the vulva's combination of keratinized (non-mucosal) and nonkeratinized (mucosal) surfaces, which may result in low cellular yield from keratinized areas and impact the ascertainment of vulvar HPV endpoints.[43] To address these challenges, our research team adhered strictly to standard operating procedures and leveraged extensive experience in vulvar swab collection for genital viral studies. Vulvar specimens were consistently collected by two experienced providers, and specimen cellularity was monitored to ensure adequate collection, thereby minimizing potential inaccuracies in our genotyping results.

Lastly, the study population was heterogeneous (e.g., sex, initial HSIL site, immunocompetence among participants), and high-grade lesion progression may vary between these groups. We purposely designed our study to combine two HPV-related diseases that are both predominantly related to HPV16 and recur in approximately 30%–50% of patients after treatment to recruit a sufficient number of participants and provide a definitive result. Furthermore, we used a dynamic randomization to optimally balance the characteristics between the active vaccine and placebo. The predominance of White participants and the middle-aged age distribution limit the generalizability of the findings to more diverse populations.

In conclusion, the 9vHPV vaccine delivered after treatment of anal or vulvar HSIL did not reduce HSIL recurrence or hrHPV DNA detection. Despite limitations, our trial contributes meaningful findings to the existing literature indicating no therapeutic effect of the current prophylactic vaccines on pre-existing HPV infection and disease. Our study underlines the importance of primary HPV prevention with prophylactic HPV vaccines in HPV-naïve persons and the need for novel therapeutic vaccines and antivirals to manage prevalent HSIL that have a high potential to recur.

AUTHOR CONTRIBUTIONS

Helen C. Stankiewicz Karita: Conceptualization; investigation; writing – original draft; methodology; visualization; writing – review and editing; project administration. **Amalia S. Magaret:** Conceptualization; methodology; data curation; investigation; validation; visualization; writing – review and editing; formal analysis; supervision. **David R. Doody:** Conceptualization; methodology; validation; visualization; writing – review and editing; formal analysis; data curation. **Jeffrey T. Schouten:** Investigation; writing – review and editing; supervision; conceptualization. **Constance Mao:** Investigation; writing – review and editing; supervision; conceptualization. **Warner K. Huh:** Investigation; writing – review and editing; supervision. **Verena S. Grieco:** Investigation; writing – review and editing; supervision; conceptualization. **Matthew A. Seymour:** Investigation; writing – review and editing; project administration. **Dana Varon:** Investigation; writing – review and editing; project administration. **Kelly H. Shrader:** Writing – review and editing; project administration; investigation. **Long Fu Xi:** Investigation; writing – review and editing; supervision; conceptualization. **Denise A. Galloway:** Supervision; writing – review

and editing; investigation; conceptualization. **Anna Wald:** Investigation; writing – review and editing; conceptualization; funding acquisition; methodology; formal analysis; project administration; supervision; resources. **Margaret M. Madeleine:** Conceptualization; investigation; funding acquisition; methodology; writing – review and editing; formal analysis; project administration; supervision; resources.

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CONFLICT OF INTEREST STATEMENT

AW receives consulting fees from Aicuris, GSK, Heidelberg Therapeutics, Innovative Molecules, and Bayer (past) and Merck (past) and research funding from Moderna, GSK, and Assembly Biosciences outside of the submitted work. DG has received research funding and honoraria from Merck for unrelated projects. All remaining authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The study protocol and informed consent received ethical approval from the Fred Hutchinson Cancer Center Institutional Review Board (FHIRB #9790). Written informed consent was obtained prior to participation in the study. The VIVA Trial is registered in ClinicalTrials.gov under registration number NCT03051516. No sex-based or racial/ethnic-based differences were present in this trial. Race, ethnicity and gender were representative and well balanced between the intervention and control arms of the trial.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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